



Institute Hall Booking System

Book halls, classrooms, labs & seminar rooms in seconds



Problem Statement

"Manual hall booking through emails is inefficient, time-consuming, and error-prone."

Manual Requests

Faculties sent booking requests manually to HOD via email

Manual Verification

HODs checked availability by hand, causing delays

Scheduling Conflicts

Managing multiple emails and conflicts was difficult

No Visibility

No centralized view of booked vs available halls

Proposed Solution

"A centralized real-time hall booking platform for institutes."



Instant Visibility

Faculty can view available halls instantly



FIFO Processing

Booking requests handled in order of arrival



One-Click Approval

HODs or Department Admins accept/reject instantly



Reduced Overhead

Drastically reduces communication overhead



Real-Time Status

Shows live booking status and event duration

Key Features

1

Department Authentication

Secure login using department emails & passwords

2

Super Admin Dashboard

Create/remove departments, halls, and admins

3

Department Admin Panel

Manage department halls & booking requests

4

Faculty/Student Dashboard

Request halls and view booking status

5

Real-Time Availability

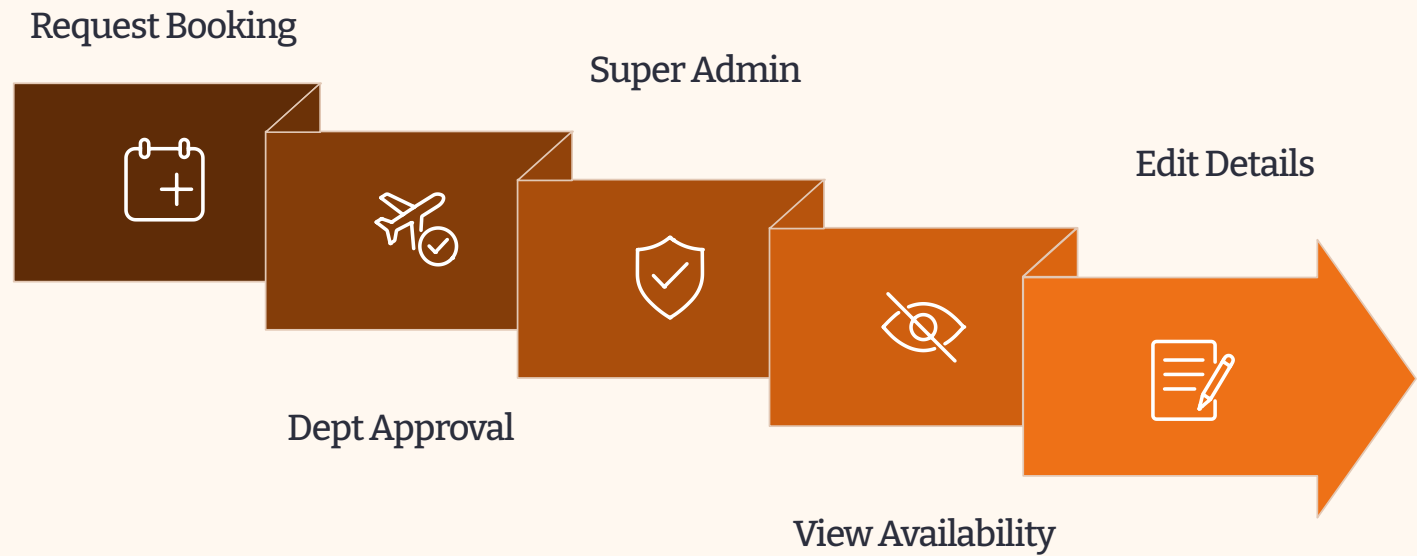
Instant display of hall availability

6

Automatic Conflict Resolution

First request gets priority automatically

Use Case Flow



The system streamlines the entire booking process from request to approval, ensuring transparency and efficiency at every step.

System Users & Roles



Super Admin

Controls all departments

- Create/Delete departments
- Manage all halls globally



Department Admin

Handles own department

- Approve/reject requests
- Manage department halls

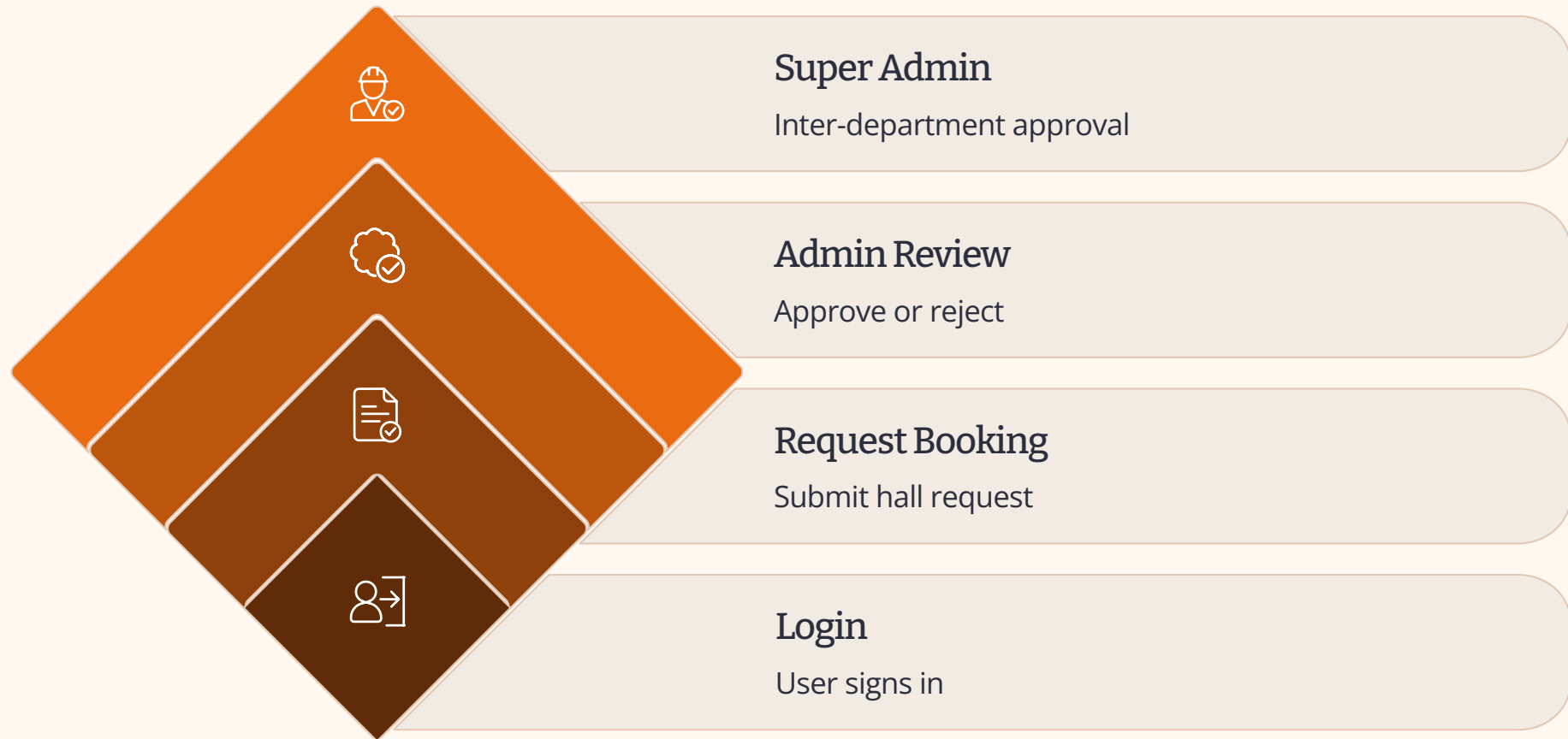


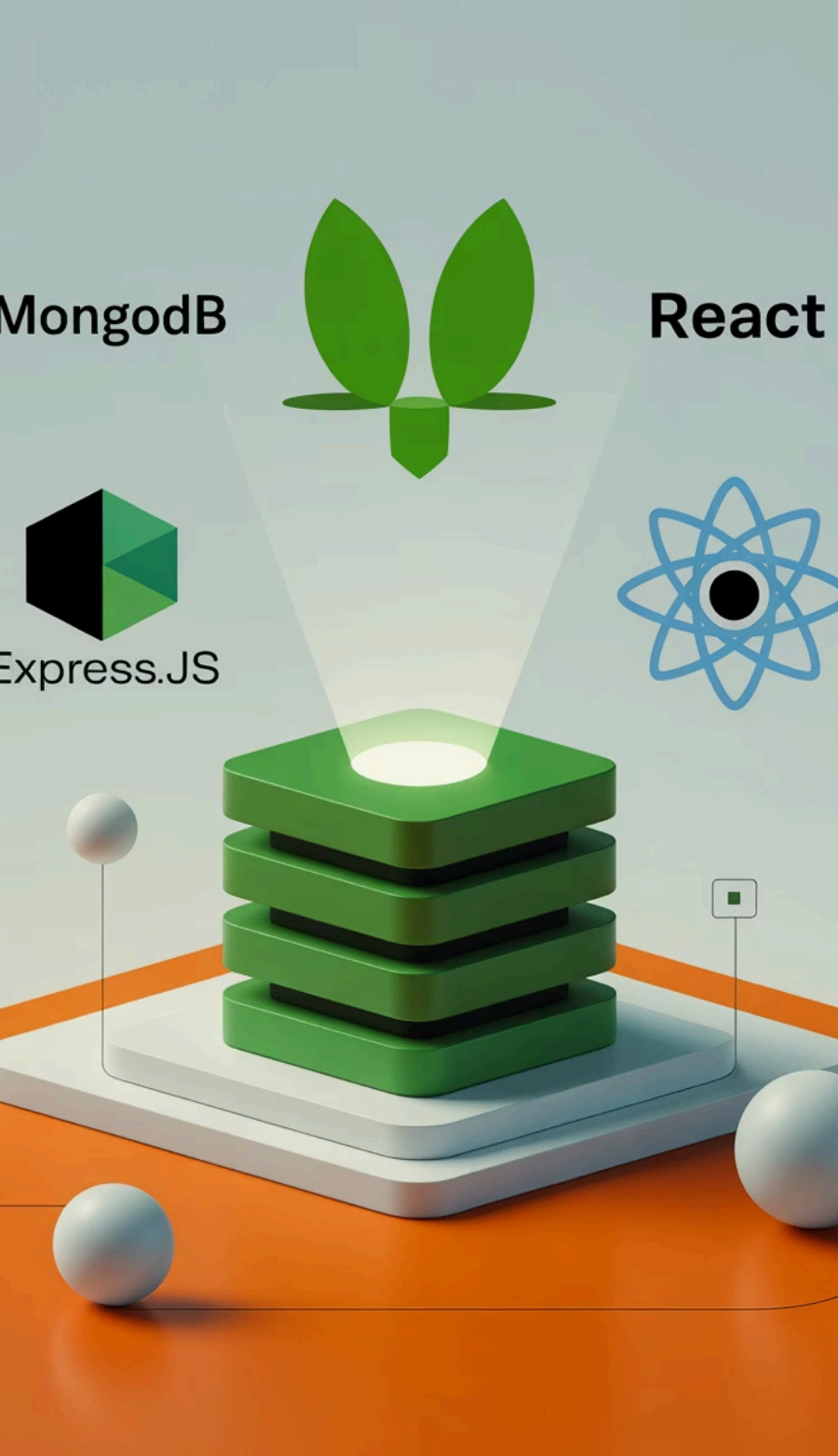
Faculty/Student

Sends hall booking requests

- Request booking
- View hall status

Workflow Diagram





System Architecture



Frontend

React.js

Components for Admin, Department, and Dashboard interfaces



Backend

Node.js + Express.js

REST APIs for CRUD operations



Database

MongoDB

Collections: Departments, Halls, Requests, Admins



Deployment

Cloud Server

Hosted on Render/Heroku or local server

Module Workflow

01

Admin Module

Manages departments, halls, and bookings across the institute

02

Department Module

Handles internal booking requests and approvals

03

Authentication Module

JWT-based secure login and access control

04

Hall Management Module

CRUD operations for hall creation and updates

05

Booking Module

Request queue management and status updates

Gaps Identified

No Real-Time Feedback

Manual process lacked instant status updates

Fragmented Coordination

No unified platform for multi-department booking

Human Delays

Overlapping bookings and scheduling errors

Poor Access Control

Lack of proper audit trails and permissions



Achievements

80%

Time Reduction

Reduced hall booking approval time significantly

100%

Email Elimination

Removed need for manual email tracking

0

Conflicts

Real-time conflict resolution eliminates double bookings



Scalable Backend

Built robust architecture with MongoDB & Express



Modern UI

Designed responsive, intuitive React interface



Future Scope



ERP Integration

Connect with institute's calendar and existing ERP systems



Real-Time Notifications

Send instant alerts via email and SMS



Auto-Reminders

Automated reminders for upcoming booked events



Analytics Dashboard

Track usage patterns and generate insights



Mobile App

Develop native mobile version using React Native

Conclusion

The Institute Hall Booking System eliminates manual scheduling inefficiencies and brings digital transparency to academic resource management. It ensures faster decisions, organized data, and smarter campus operations.